Andrea Muolo

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EXPERIENCE

.NET Software Engineer and Business Analyst

Systemorph AG

Dec 2021 - Mar 2024 Zürich, Switzerland

Developer of the Systemorph Cloud platform and client-tailored data-driven scalable web applications for the insurance industry sector with a focus on observability and auditability.

Experience with modern cloud architecture principles and patterns. Researched, designed and implemented concurrency formalism founded on the actor model and microservices.

Led POCs and responsible for meeting internal and external stakeholders, discussing business specifications, from data modelling, to business logic (e.g. actuarial methodologies, accounting standards), business processes, UX design.

Organize and conduct presentations to clients, on-site trainings, and workshops.

Developer and tester of in-house C# packages and concept applications to showcase our technology and foster our community. Tech advocate.

Postdoctoral Research Fellow

Hebrew University of Jerusalem

Oct 2019 - Nov 2021 Jerusalem, Israel

Exact Factorization techniques in molecular physics. Algebraic manipulation of exact factorization equation and numerical calculations to tackle non-adiabatic systems. Genuine predictions on the Berry phase. Methods developed in C++20.

Research and Teaching Assistant

ETH Zürich

Dec 2018 - Sept 2019 Zürich, Switzerland

Many-body Dirac Equation and relativistic quantum mechanics. Novel analytical and numerical solutions to integrals to enable symmetry projected optimization.

Method development: Pre-Born–Oppenheimer and Tensor Networks algorithm.

Rich experience in concurrent and distributed computing using C++17.

EDUCATION

PhD in Computational Molecular Physics

ETH Zürich

Nov 2014 – Dec 2018 Zürich, Switzerland

Explicitly Correlated Gaussians and the Quantum Few-Body Problem.

BSc and MSc in Physical Chemistry

University of Pavia and University of Cambridge

Sept 2009 – Sept 2014 Pavia, Italy – Cambridge UK

MSc : A classical dynamic that conserves the quantum Boltzmann distribution.

BSc: Molecular Memories: rotaxane- and catenane-based molecular devices.

Second level MSc at Scuola Universitaria Superiore (IUSS)

MOST PROUD OF

Swiss National Science Foundation (SNSF) Fellowship
Research Grant, October 2019 - Nov 2021

Project Manager - from Jul 2023
Define scope, plan and organize the execution of the POC

TECH STACKS

2+ years

- .NET, C#, WPF, Razor, Blazor ●●●●●
- Nuget, SignalR, ML.NET, SQL, EF, DI ●●●●●
- WPF, Razor, Blazor, MVC, MVVM ●●●●●
- Javascript, AJAX, HTML5, CSS ●●●●○
- Kubernetes, Docker, CI/CD ●●●●○

7+ years

- C++20, OpenMP, MPI, SIMD, HPC ●●●●●
- Armadillo, Boost, Eigen, Intel MKL ●●●●●
- Python, OpenACC, CUDA ●●●●○
- Eclipse, Visual Studio (+ReSharper) ●●●●●
- Git, Mathematica, LaTex ●●●●●

STRENGTHS

Algorithm Optimization | Strong Math Foundation | Concurrent Computation | Distributed Computing | Quantum Computing | Software Design | Scrum and agile framework | Cloud Architectures

Data Modelling | Business Process | UX Design | Actuarial Methods | Accounting Standards IFRS

Fast Learning | Scientific Writing | Teaching | Public Speaking | Cross Domain Knowledge

LANGUAGES

German (A2) **English** (fluent) **Italian** (native)

ABOUT ME

SELECTION OF PUBLICATIONS

J. Chem. Theory Comput., 18, 234 (2022)
Quantum Proton Effects from Density Matrix Renormalization Group Calculations

Phys. Rev. A, 102, 022803 (2020)
Analytically projected, rotationally symmetric, explicitly correlated Gaussians with one-axis shifted centers

J. Chem. Phys., 142, 134103 (2015)
Boltzmann-conserving classical dynamics in quantum time-correlation functions: “Matsubara dynamics”